Lab 9 – Logistic and Poisson Regression Analysis

Logistic Regression

Model accuracy:

The logistic model had an accuracy of 68.42%.

Lowest Precision:

The model with the lowest precision was category 1 at 0.67, compared with 0.69 at category 0.

Both categories have at least 30 observations in the classification report:

Both categories did not have at least 30 observations in the classification report. Category 0 had 25 observations, while category 1 had 13 observations.

Pseudo-R^2:

The pseudo-R^2 value was 0.0625, which indicates that the model didn't really improve upon a model without any predictors. What I take from this is that some predictors like smoking and hypertension alone, do not strongly predict low birth weight.

Model Accuracy: 68.42%

Classification Report:

	precision	recall	f1-score	support
0	0.69	0.96	0.80	25
1	0.67	0.15	0.25	13
accuracy			0.68	38
macro avg	0.68	0.56	0.53	38
weighted avg	0.68	0.68	0.61	38

Confusion Matrix:

[[24 1], [11 2]]

Pseudo R^2: 0.0625

Poisson Regression

Four Statistically Significant Predictors:

The four predictors I identified statistically significant were depression, self-care, independent living, and loneliness.

Which predictor is most significant:

Depression was the most statistically significant predictor, with a z-score of 5.800 and a p value below 0.001. However, self-care had the largest coefficient in absolute value at -0.0608.

Statistically significant predictors with VIF greater than 10:

Both self-care and independent living had a VIF above 10, Self-care had a VIF of 30.51, and independent living had a VIF of 30.95.

Meaning of pseudo-R^2:

The pseudo-R^2 value was 0.04356, with such a low value, this tells me that this model also didn't improve much over a model without predictors.

Current function value: 2.347799

Iterations 5

Poisson Regression Results

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Dep. Variable: mhlth_crudeprev No. Observations: 2299

Model: Poisson Df Residuals: 2287

Method: MLE Df Model: 11

Date: Sun, 25 Oct 2026 Pseudo R-squ.: 0.04356

Time: 23:39:09 Log-Likelihood: -5397.6

converged: True LL-Null: -5643.4

Covariance Type: nonrobust LLR p-value: 1.908e-98

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	coef	std err	z	P>|z|	[0.025	0.975]
const	1.7032	0.172	9.888	0.000	1.366	2.041
binge_crudeprev	0.0032	0.003	0.954	0.340	-0.003	0.010
checkup_crudeprev	-0.0004	0.002	-0.192	0.848	-0.004	0.004
csmoking_crudeprev	0.0043	0.004	1.100	0.271	-0.003	0.012
depression_crudeprev	0.0137	0.002	5.800	0.000	0.009	0.018
obesity_crudeprev	0.0001	0.002	0.076	0.940	-0.003	0.004
sleep_crudeprev	0.0024	0.002	1.059	0.289	-0.002	0.007
selfcare_crudeprev	-0.0608	0.022	-2.708	0.007	-0.105	-0.017

indeplive_crudeprev	0.0522	0.014	3.759	0.000	0.025	0.079
loneliness_crudeprev	0.0113	0.003	3.717	0.000	0.005	0.017
foodinsecu_crudeprev	-0.0083	0.007	-1.247	0.212	-0.021	0.005
housinsecu_crudeprev	0.0150	0.008	1.786	0.074	-0.001	0.032

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Variance Inflation Factors:

	Predictor	VIF
0	binge_crudeprev	2.528842
1	checkup_crudeprev	1.750444
2	csmoking_crudeprev	6.934794
3	depression_crudeprev	2.216376
4	obesity_crudeprev	2.573017
5	sleep_crudeprev	3.209334
6	selfcare_crudeprev	30.514413
7	indeplive_crudeprev	30.953309
8	loneliness_crudeprev	3.432565
9	foodinsecu_crudeprev	62.830787
10	housinsecu_crudeprev	46.839954